MJCET

Project: Take-Out Robot

Lastest Update on: 25-03-2022

Assembly No: TR-SA2

Assembly: X-AXIS

Item No	Description	Drawing Number	CAD Part Name	Material Standard	Size	Weight (kg)	Qty/Set	Manufacturer /Vendor	Manufacturer Part No	Notes
2.1	X-Axis Beam Channel		2.1_X-Beam Channel	Aluminium Alloy (AL 6063 T6)	Length 2700mm	30.40	1	Sica	PG80160H	
2.2	X-Axis Guide Rail		2.2_X-Guide Rail			28	4	Hiwin	RGR20R2700H	
2.3	Guide Block		2.3_X-Guide Block			2.4	4	Hiwin	RGH20CAZ0H	
2.4	Stopper	TR-SA2-010	2.18 X Stopper	SS304	As per drawing	0.04	8			
2.5	Allen Bolts		ISO 4762 M5 x 20 - 20N	SS	M5 x 20mm		2			
2.6	Allen Bolts		ISO 4762 M5 x 35 - 22N	SS	M5 x 35mm		12			
2.7	Allen Bolts		ISO 4762 M5 x 40 - 22N	SS	M5 x 40mm		4			
2.8	Allen Bolts		ISO 4762 M5 x 30 - 22N	SS	M5 x 30mm		12			
2.9	Allen Bolts		ISO 4762 M4 x 16 - 16N	SS	M4 x 16mm		6			
2.10	Allen Bolts		ISO 4762 M5 x 16 - 16N	SS	M5 x 16mm		12			
2.11	Allen Bolts		ISO 4762 M6 x 20 - 20N	SS	M6 x 20mm		4			
2.12	Allen Bolts		ISO 4762 M8 x 16 - 16N	SS	M8 x 16mm		2			
2.13	Allen Bolts		ISO 4762 M8 x 20 - 20N	SS	M8 x 20mm		2			
2.14	Allen Bolts		ISO 4762 M5 x 20 - 20N	SS	M5 x 20mm		164			
2.15	Allen Bolts		ISO 4762 M6 x 16 - 16N	SS	M6 x 16mm		4			
2.16	Allen Bolts		ISO 4762 M3 x 10 - 10N	SS	M3 x 10mm		4			
2.17	Allen Bolts		ISO 4762 M6 x 30 - 30N	SS	M6 x 30mm		8			
2.18	X PS Holder	TR-SA2-011	2.19 X Proximity Sensor Holder	SS304	As per drawing	0.08	2			
2.19	Proximity sensor		E.6 Proximiting Sensor PNP Flush N-O				2	Omron	E2A-M18KS08-WP-B1 2M	
2.20	Washer		ISO 10673-5.5-S		M5		4			
2.21	Spacer	TR-SA2-008	2.12_X-Spacer Straight	MS	As per drawing	1.2	1			
2.22	L shape spacer	TR-SA2-009	2.13_X-Spacer L-type	MS	As per drawing	1.5	1			
2.23	Timing Belt Clamp Plate		2.7_X-Clamp Plate for Timing Belt	Clear Ionized-A6N01-T5		0.07	1		TBCK-S5M250	
2.24	Beam Side Plate	TR-SA2-001	2.16_X-Channel Side Plate	Aluminium 6063	As per drawing	0.17	2			
2.25	Motor Clamp	TR-SA2-005	2.17 X-motor Clamp	SS304	As per drawing	0.36	1			
2.26	AC Servo Motor with encoder included		hg-kn73j			12	1	Mitsubishi	HG-KN 73(B) J	
2.27	Motor plate	TR-SA2-006	2.15 X Motor Plate	SS 304	As per drawing	0.95	1			
2.28	Nylon Lock Nut		ISO 10511-M6-N	SS 302			4			
2.29	Plate Idler Pulley	TR-SA2-007	2.9_X-Plate Idler Pulley	MS	As per drawing	0.13	1			
2.30	Cantilever Shafts - Shouldered - Lock Nut		2.8_X-Cantiliver Shaft	Steel	Dia-10mm	0.1	1	Misumi	FXFC15-12-F25-M8	
2.31	Timing Pulley		HTPA26S5M250_A_C19	[Steel S45C Equivalent	19mm ID Width 25mm	0.1	1	Misumi	HTPA26S5M250-AC19	
2.32	Idler Pulley		C_AHTFW26_S5M250_10	Aluminium Alloy	10mm ID Width 25mm	0.07	1	Misumi	CAHTFW26-S5M250-10	

2.33	Timing Belt		x timing belt^TR-SA-2	Rubber Belt	Length 5572.5 mm Width 25mm	0.57	1	Misumi	HTBO6-S5M250-1095	
2.34	Carriage Assembly	TR-SA2-A1	A1_X-Carriage	MS	As per drawing		1			
2.35	Allen Bolts		ISO 4762 M4 x 20 - 20N	SS	M4 x 20mm		8			